Chapter 1.4 Integration

Rick Durrett Probability: Theory and Examples

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Built from the local Chapter 1 LLMviki flash cards.

Roadmap for 1.4

Learning goal

Understand the Lebesgue integral construction well enough to know why approximation arguments work.

1. Simple functions and canonical level-set sums.
2. Nonnegative integral as lower approximation.
3. Signed integrability through positive and negative parts.
4. Linearity, monotonicity, and a.e. replacement.
5. Approximation exercises: null sets, dyadic bins, L^1 density, and Riemann–Lebesgue.

Simple Functions

Definition

A measurable function ϕ is simple if it takes finitely many values. In canonical form,

$$\phi = \sum_{k=1}^{n} a_k \mathbf{1}_{A_k},$$

where the A_k are measurable level sets.

Integral

For $\phi \geq 0$,

$$\int \phi \, d\mu = \sum_{k=1}^{n} a_k \mu(A_k).$$

LLMviki: `Simple functions; ID: durrett_ch1_simple_functions`

Simple-Function Proof Template

How to prove integral identities

Prove the statement first for indicators, then for nonnegative simple functions, then pass to limits.

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Integral of a Nonnegative Function

Definition

If $f \geq 0$ is measurable, define

$$\int f \, d\mu = \sup \left\{ \int \phi \, d\mu : 0 \leq \phi \leq f, \ \phi \text{ simple} \right\}.$$

Pitfall

The value may be $+\infty$. Do not subtract two infinite nonnegative integrals.

LLMviki: `Integral of a nonnegative function; ID: durrett_ch1_nonnegative_integral`

Monotone Approximation Idea

Proof idea

The definition makes lower approximation the natural language for nonnegative integrals.

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Integrable Signed Functions

Definition

For a real measurable f, write

$$f^+ = \max(f, 0), \qquad f^- = \max(-f, 0).$$

Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

Integrability

We say f is integrable if

$$\int f^+ \, d\mu < \infty \quad \text{and} \quad \int f^- \, d\mu < \infty,$$

equivalently $\int |f| \, d\mu < \infty$.

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Signed Integral: How to Define It

Definition

If f is integrable, set

$$\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu.$$

LLMviki: `Integrable signed functions; ID: durrett_ch1_integrable_function`

Linearity and Order Properties

Theorem

For integrable functions, the integral is linear and monotone:

$$\int (af + bg)\, d\mu = a \int f\, d\mu + b \int g\, d\mu,$$

and if $f \leq g$ a.e., then

$$\int f\, d\mu \leq \int g\, d\mu.$$

If $f = g$ a.e., then $\int f\, d\mu = \int g\, d\mu$.

LLMviki: `Linearity and order properties of the integral; ID: durrett_ch1_integral_linearity`

Linearity: How to Prove It

LLMviki: `Linearity and order properties of the integral; ID: durrett_ch1_integral_linearity`

Exercise 1.4.1

Show that if $f \geq 0$ and

$$\int f \, d\mu = 0,$$

then $f = 0$ almost everywhere.

LLMviki: `Zero integral of nonnegative function implies zero a.e.; ID: exercise_method_zero_integral_nonnegative`

Exercise 1.4.1: Proof Skeleton

LLMviki: `Zero integral of nonnegative function implies zero a.e.; ID: exercise_method_zero_integral_nonnegative`

Exercise 1.4.2

Let $f \geq 0$ and

$$E_{n,m} = \left\{ x : \frac{m}{2^n} \leq f(x) < \frac{m+1}{2^n} \right\}.$$

Show that, as $n \to \infty$,

$$\sum_{m=1}^{\infty} \frac{m}{2^n} \mu(E_{n,m}) \uparrow \int f \, d\mu.$$

LLMviki: `Dyadic lower simple approximations; ID: exercise_method_dyadic_lower_integral_approximation`

Exercise 1.4.2: Proof Skeleton

LLMviki: `Dyadic lower simple approximations; ID: exercise_method_dyadic_lower_integral_approximation`

Exercise 1.4.3

Let g be an integrable function on $\mathbb{R}$ and let $\varepsilon > 0$. Show, using the definition of the integral and interval approximation of measurable sets, that g can be approximated in L^1 by a continuous function. More explicitly:

1. approximate g by a simple function ϕ with $\int |g - \phi| \, dx < \varepsilon$;
2. approximate the sets in ϕ by finite unions of intervals to obtain a step function q with $\int |\phi - q| \, dx < \varepsilon$;
3. round the corners of q to obtain a continuous function r with $\int |q - r| \, dx < \varepsilon$.

LLMviki: `L1 approximation by continuous functions; ID: exercise_method_l1_continuous_approximation`

Exercise 1.4.3: Proof Skeleton

LLMviki: `L1 approximation by continuous functions; ID: exercise_method_l1_continuous_approximation`

Exercise 1.4.4

Prove the Riemann–Lebesgue lemma: if g is integrable, then

$$\lim_{n \to \infty} \int_{\mathbb{R}} g(x) \cos(nx)\, dx = 0.$$

Hint: first prove it for step functions, then use the previous exercise.

LLMviki: `Riemann-Lebesgue by step-function reduction; ID: exercise_method_riemann_lebesgue_step_reduction`

Exercise 1.4.4: Proof Skeleton

LLMviki: Riemann-Lebesgue by step-function reduction; ID: exercise_method_riemann_lebesgue_step_reduction